SPPA: A Deterministic Runtime Contract for Semantic Primitive Proxy Actors in UAV Digital Twins

Authors omitted for review

Abstract

Object-level UAV telemetry is not directly renderable: a detector may provide a class label, confidence, track identifier, bounding region, and approximate world pose, but no mesh. This paper introduces Semantic Primitive Proxy Assembly (SPPA), a deterministic semantic actor contract that converts sparse object telemetry into low-polygon, role-labeled primitive parts when curated 3D assets are unavailable, too expensive, or semantically mismatched.

SPPA separates runtime representation from visual asset generation. Reviewed archetype recipes map detector labels to part descriptors such as cab, cargo, wheels, trunk, canopy, torso, or fallback volume; runtime packets then update pose, confidence, scale evidence, and material tags without regenerating mesh topology when the part layout is unchanged. The current implementation includes a finite-archetype prototype generator, explicit unknown-label fallback, a `SPPA-DESC-0.2`/`SPPA-UPD-0.2` descriptor/update contract, and an optional Unreal Engine 5.7 backend switchable against curated assets using the same obstacle input. SPPA does not promise arbitrary perfect reconstruction; it converts imperfect semantic evidence into lightweight, controllable, safety-oriented 3D proxies. Local measurements show millisecond-scale descriptor creation in the typical case, cheap update packets, limited packaged desktop feasibility, and clear limits where curated assets or neural generators are better choices. The paper does not claim dense VR readiness, operator benefit, live-flight validation, or a universal bandwidth advantage.

1 Introduction

Semantic-telemetry UAV digital twins replace part of the continuous video stream with object-level messages. A UAV-side perception stack can report object tracks, classes, confidence, positions, and approximate geometry. The ground station then has to represent those objects in an Unreal/Cesium digital twin. That architectural choice creates a rendering problem that is separate from the flight system itself: object-level telemetry is not a mesh.

Curated 3D assets are useful when the class is known, the asset library is complete, and runtime loading is acceptable. Those assumptions are weak for field UAV operation. Long-tail labels, ambiguous detections, partial views, occlusion, motion blur, and changing object proportions make a fixed asset library brittle. Neural image-to-3D and text-to-3D methods address a different objective: visual asset generation. They can produce richer geometry, but they depend on usable crops or prompts, GPU inference, and mesh outputs that may be too dense for a portable flight laptop that is also running Unreal, telemetry, video, and tracking under a shared rendering budget.

SPPA proposes a lower-fidelity representation point. Instead of reconstructing the real mesh, it compiles a detection into a semantic part graph made from primitive boxes, cylinders, ellipsoids, cones, and tori. A vehicle becomes a cab, body/cargo region, wheels, and optional top details. A quadruped becomes a torso, head, and legs. A tree becomes a trunk and canopy volumes. Unknown or unsupported labels are not hallucinated into specific objects; they fall back to a generic uncertainty-aware display volume with explicit uncertainty metadata.

The scientific question is therefore not whether SPPA looks better than neural 3D generation. It should not. The question is whether a deterministic, track-persistent, updateable primitive actor is a useful runtime representation for UAV digital twins when the available input is a semantic detection rather than a clean image crop or a curated mesh.

2 Contribution and Scope

The main contribution is a semantic telemetry-to-actor contract, not a new UAV flight pipeline. SPPA consumes the same object input as a conventional Unreal asset-spawning path and produces a switchable alternative representation: curated/static mesh assets or generated semantic primitive actors. This separation keeps the claim narrow: this paper concerns only the object-representation layer that turns semantic detections into runtime actors.

The contribution is divided into four claims:

- **Semantic normalization.** Approximate detector evidence is not copied directly into the scene. It is routed through a SPPA normalizer that maps labels such as `vehicle`, `agricultural vehicle`, or `electric pylon` to reviewed proxy families or conservative fallback.
- **Contract.** A semantic descriptor/update format separates actor creation from pose, material, uncertainty, and matching-topology shape updates.
- **Runtime role.** The same obstacle input can select curated assets or generated primitive actors, so SPPA is an optional representation for uncovered or mismatched labels rather than a replacement for good assets.
- **Evidence boundary.** Current evidence is limited to local resolver, descriptor, synthetic geometry, and Unreal packaged tests; real UAV masks, dense VR, operator benefit, and measured radio-link advantage remain outside the supported claim.

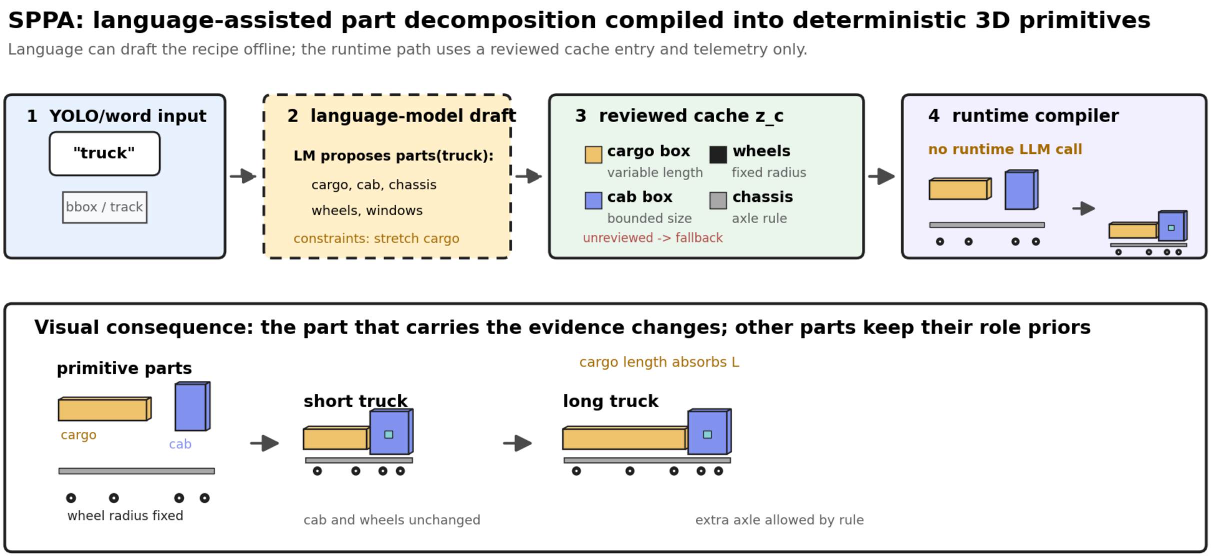

Figure 1: SPPA semantic-label-to-parts-to-3D mechanism for the truck archetype. A language model, ontology tool, or human author may draft candidate part roles offline, but runtime execution uses only a reviewed cache entry z_c. The figure shows the proposed chain explicitly: an input label or word activates an offline draft, the draft is reduced to reviewed part roles and bounded update rules, and runtime compiles those roles into primitive geometry. The short and long proxy variants illustrate the implemented role-specific rule: cargo length changes while cab and wheel priors remain bounded, so the object is not represented by root-scaling a single box.

Claim boundary: Figure 1 is not a claim of photorealistic reconstruction or universally correct geometry; SPPA is only claimed here as a bounded, reviewed semantic-proxy compiler.

Figure 1 states the central representation claim visually: SPPA is a reviewed semantic recipe compiler, not arbitrary word-to-3D generation and not a primitive-shape benchmark. The language component is limited to offline decomposition drafting and ontology expansion. The implemented runtime path starts after that cache exists: a detector label and telemetry state select a reviewed archetype, fill bounded part parameters, and expose explicit fallback behavior when the label is unsupported. For vehicles, the important distinction is part-parametric adaptation rather than global mesh or root-scale deformation.

3 Language Model Use and Geometry Path

The current SPPA implementation does not call a language model in the runtime path. The implemented resolver contract records `resolver_source` as a static keyword ontology and marks `runtime_llm_used` as false; the versioned recipe manifest also disallows runtime LLM use. The executable system described in this paper should therefore not be read as a live LLM that answers questions such as "is this a mammal?" or "how many legs should it have?" before constructing a mesh. Those facts are already encoded in reviewed archetype recipes. If an LLM is used at all, it is only an offline authoring aid that can propose candidate part decompositions; those candidates must be reviewed, versioned, regression tested, and cached before runtime use. The present artifacts record that offline assistance category, but they do not record a specific LLM provider, model name, prompt, or output hash, so the paper makes no claim about a particular language model.

Two input modes are therefore distinct. A detector-driven input contains a class label plus confidence, track state, bounding region, optional mask, pose, and possibly calibrated scale. SPPA may use that evidence to choose a family, derive an axial footprint, update pose, or adapt dimensions. A tag-only input contains only a word or phrase; it can select a reviewed recipe and default dimension priors, but it cannot provide real silhouette, pose, color, or metric scale evidence. This distinction is important for the benchmark figures: a text-only prompt activates a recipe, whereas a detected object can carry measurement evidence.

Runtime execution is a deterministic compilation pipeline:

1. A detector label or tag is normalized to a SPPA tag and runtime label. For example, labels such as `electric pylon`, `agricultural vehicle`, or `vehicle` are mapped to reviewed semantic families such as vertical structure, farm vehicle, heavy vehicle, or generic vehicle when the evidence supports only that level.

2. The generator resolves the normalized label through exact-class rules, keyword-archetype rules, verifier-gated open-label proxy recipes, or explicit unknown fallback.

3. Metric dimensions come from supplied dimensions, calibrated footprint evidence, or bounded archetype priors. Without calibration, a mask remains image-space evidence and does not by itself become metric geometry.

4. The selected recipe emits role-labeled primitives, such as body, cab, cargo, wheel, torso, head, leg, trunk, canopy, mast, crossbar, or fallback volume. When parts are semantically attached, the recipe also emits explicit lightweight connector cylinders between contact points, for example bicycle frame tubes, rider-to-bike limbs, tower braces, or tractor-trailer hitches. Each primitive records type, local transform, scale, material role, and uncertainty metadata.

5. The descriptor is consumed by Unreal, which creates primitive components once per track and archetype and then applies pose, confidence, material, or compatible shape updates without rerunning a generative model.

For a cow-like object, SPPA does not infer animal anatomy from first principles at runtime. A label such as `cow` or a conservative quadruped family selects a reviewed recipe whose topology already contains torso, head, and legs. For a tractor or tractor-trailer, the system need not detect the exact taxonomic class to be useful: a detector output such as `vehicle`, `agricultural vehicle`, `truck`, or `trailer` can be normalized to a farm, heavy, or generic vehicle proxy. If the label is weak or adversarial, SPPA should prefer a conservative fallback volume over inventing a specific object. In the current implementation, a small set of useful UAV open-label cases such as `forklift`, `traffic cone`, `water tank`, `shipping container`, `drone`, and `hay bale` may pass through a deterministic recipe verifier before they are accepted as generated proxy recipes; otherwise fallback remains the runtime behavior.

This is also the honest novelty boundary. Primitive composition, CSG, shape programs, procedural modeling, and text-to-CAD are mature research areas [2, 35, 43, 57, 34, 24, 15, 16, 18, 33]. SPPA's contribution is not that boxes and cylinders can describe objects. Its contribution is a bounded UAV-runtime contract that converts imperfect semantic evidence into lightweight, controllable, updateable, and fallback-safe 3D proxy actors.

4 Representation Design Space

The relevant comparison is not a single "3D representation" baseline. A UAV digital twin can visualize a semantic detection through several families of representations, each making different assumptions about input evidence, runtime budget, geometry, and operator interpretation. Geometry-only displays such as boxes, ellipsoids, or billboards remain useful lower-bound controls, but they are not the scientific competitor. SPPA targets the gap between a generic box and a curated asset: a role-labeled primitive actor when only sparse semantic telemetry is available.

Table 1 states the resulting deployment policy. It deliberately gives priority to curated assets and mapping geometry when those are the correct tools.

Table 1: Operational representation policy for semantic detections.

Situation	Preferred representation	Reason	Boundary
Matched project asset exists	Curated asset backend	best appearance and lower measured cost in the cow/tower subset	asset library must match class and scale
Known semantic family, no asset	SPPA primitive actor	persistent, local, role-labeled 3D proxy from label/pose input	finite reviewed archetypes
Unknown or adversarial label	SPPA fallback volume	avoids inventing a specific object from weak semantics	not visually specific
Clean crop plus offline budget	Image-to-3D or asset authoring	richer mesh can be generated outside the flight update loop	no per-frame track contract
Planning or occupancy task	Occupancy / ESDF / map layer	represents navigable space and collision evidence	not a role-labeled display actor

5 Related Work

SPPA overlaps with several mature research areas, but it does not replace them. Text-to-3D and image-to-3D generation methods such as DreamFusion, Magic3D, Point-E, Shap-E, TripoSR, TripoSG/Tripo P1, InstantMesh, Stable Fast 3D, SPAR3D, LGM, CRM, Unique3D, Direct3D-S2, PartCrafter, Pixal3D, TRELLIS/TRELLIS.2, Hunyuan3D 2.x, and commercial systems such as Rodin Gen-2.5 target visual asset generation or reconstruction [30, 21, 25, 17, 42, 45, 52, 36, 37, 38, 39, 46, 47, 48, 22, 19, 51, 50, 56, 40, 14]. Their strength is fidelity when suitable visual or prompt evidence is available. SPPA targets a different runtime point: deterministic actor generation from sparse semantic telemetry, low polygon counts, and cheap per-frame updates. Recent benchmark work also highlights why a real image-to-3D ranking needs a shared reference set, metrics, and part/structure evaluation rather than a local figure assembled from heterogeneous inputs [41, 53].

For this reason, the artifact package includes an explicit SOTA-protocol readiness manifest and verifier. In the current artifact state that verifier keeps the comparative figure classified as an input-alignment audit, not as an image-to-3D SOTA leaderboard.

Monocular and open-vocabulary 3D detection estimate 3D boxes or object state from images [3, 54]. Animal pose and animal reconstruction methods estimate detailed skeletons or category-specific 3D shape [55, 20, 13]. These methods can provide stronger evidence to SPPA, but SPPA itself is a rendering/backend layer, not a detector or pose estimator.

Primitive representations have a long history. Marr's volumetric primitives, recognition-by-components, superquadrics, learned part decomposition, and recent superquadric scene decomposition all show that structured primitive geometry can encode meaningful shape abstractions [23, 2, 35, 29, 43, 10]. Neural primitive and CSG parsers show that networks can infer compact primitive or boolean programs from shape evidence [57, 34]. Procedural modeling, shape grammars, and learned shape programs show how compact programs can generate or abstract structured objects [24, 15, 16]. Text-to-CAD and point-cloud-to-code systems extend this program-generation line to natural-language CAD prompts and LLM-assisted CAD reconstruction [18, 33]. SPPA uses these ideas operationally, but the claimed novelty is the bounded, track-updateable semantic telemetry contract inside an Unreal UAV digital twin.

Graphics systems already provide level-of-detail, impostors, billboards, streaming tiles, and virtualized geometry [4, 5, 8, 28]. 3D and dynamic scene graphs maintain semantic world structure [1, 32, 49]. Occupancy grids, OctoMap, ESDF, and TSDF maps support planning and mapping [12, 27]. SPPA should be read as a display actor representation that can consume such upstream evidence, not as a mapping substrate.

Finally, any operator-facing claim must be evaluated with human-factors methods. SAGAT, situation-awareness theory, NASA-TLX, and uncertainty-display literature are relevant to future SPPA validation [6, 7, 11]. The present paper contains no pilot study.

6 SPPA Contract

At time t, SPPA consumes an object state

$$x_t = \{i, \ell, c, b_t, m_t, \xi_t, s_t, \Sigma_t, h_t\},$$

where i is a track identifier, ℓ a detector label, c confidence, b_t a bounding region, m_t an optional mask or footprint, ξ_t a world pose estimate, s_t optional metric dimensions, Σ_t uncertainty, and h_t the track history. The resolver maps ℓ to one of three cases:

- `exact_class`: implemented label with a fixed archetype;
- `keyword_archetype`: out-of-template label mapped to a finite reviewed archetype, such as vehicle or vertical structure;
- `fallback_unknown_label`: unknown label rendered as a generic uncertainty-aware volume.

This is deliberately not universal word-to-3D generation. A language model may assist offline ontology expansion, but runtime resolution must remain bounded, reviewed, cached, and able to reject over-specific mappings. Figure 1 makes this boundary explicit: language-model support belongs to offline archetype authoring or review, while the runtime path maps a detector label to a reviewed archetype, generic fallback volume, descriptor parts, and Unreal components. The SPPA observation path makes the metric-evidence boundary explicit through a `SPPA-OBS-0.1` observation record. It stores detector evidence, camera and flight geometry, a projected local footprint when available, a metric dimension proposal, and a covariance-style uncertainty envelope. Unless the record marks telemetry and metric ground truth as measured, the resulting pose and scale are treated as scenario-relative evidence for a proxy contract rather than as flight-localization truth. The current implementation includes a versioned archetype-recipe contract with 15 reviewed archetypes and 64 exact/keyword/fallback label checks. That contract verifies resolver status, allowed runtime archetype, required semantic material roles, and disabled runtime LLM use; it does not validate geometric correctness for arbitrary open-vocabulary labels. A separate offline recipe-review gate now tests five candidate mappings before caching: one accepted reviewed-family mapping, one fallback-only mapping, and three rejected mappings for visual artifact context, unreviewed topology/material roles, or attempted runtime LLM use. This is an implementation guard, not proof of open-vocabulary coverage.

6.1 Part Graph

Each archetype a defines a part graph

$$G_a = (V_a, E_a), \quad v_k = (p_k, r_k, \tau_k, \rho_k),$$

where p_k is a primitive type, r_k is its role, τ_k its transform relative to the actor root, and ρ_k role-specific parameters such as dimension priors, color/material role, and confidence tags. The current implementation uses boxes, cylinders, ellipsoids, cones, and tori.

Vehicle adaptation is the clearest implemented case. A long truck and a short truck should not be produced by globally scaling the same box. In the current vehicle grammar, cab and wheel dimensions remain bounded by role priors, while the cargo/body span and axle placement absorb length changes. In the synthetic truck check, two descriptors with equal width/height and different length keep cab scale fixed at $[1.656, 2.070, 1.952]$ m and wheel scale fixed at $[0.392, 0.094]$ m, while cargo/body length changes from 2.263 m to 5.188 m. This is an internal part-invariance diagnostic, not a visual comparison against competing 3D generation methods and not proof that SPPA recovers true truck morphology from UAV imagery. The corresponding component views are kept in the artifact log rather than used as main-paper comparative evidence.

6.2 Pose, Yaw, and Updates

The descriptor separates actor pose ξ_t from part parameters q_t. Pose changes are per-frame updates; topology regeneration is avoided unless the class/archetype changes or new evidence crosses a configured threshold. A typical update packet therefore changes only root transform, confidence, material/uncertainty tags, or a small set of shape parameters.

Yaw ambiguity is handled explicitly. A top-down footprint or PCA orientation gives an axial orientation modulo π, not necessarily a signed heading modulo 2π. Signed yaw requires velocity, asymmetry, multi-frame evidence, or an external heading source. If no reliable sign exists, the descriptor sets `yaw_ambiguous=true`. The exact thresholds and weights in the fitting objective are exposed as versioned protocol parameters in the current implementation, not calibrated constants.

When a mask polygon is available, the descriptor records both the axis-aligned mask box and an oriented image-space footprint from PCA projections. This is used as footprint evidence and axial yaw evidence. If explicit metric dimensions or an explicit image-to-ground scale are supplied, the current generator can convert the calibrated footprint to metric length/width and pass those dimensions to the part-parametric builder. Without such calibration, the mask remains image-space evidence only and does not change metric geometry.

7 Implementation

The prototype has three layers. The first is a Python OBJ/MTL prototype generator for a finite set of explicit and keyword-resolved archetypes, including vehicles, vulnerable road users, vegetation, vertical structures, generic quadrupeds, and unknown fallback volumes. This path is useful for visual inspection and lightweight local timing only.

The second layer emits versioned descriptors and update packets. A `SPPA-DESC-0.2` record stores resolver source, match type, fallback reason, scale source, axis-aligned and oriented mask footprint evidence, yaw ambiguity, shape confidence, material source, optional observed-color evidence, part list, primitive counts, and scheduler metadata. A `SPPA-UPD-0.2` record separates create/regenerate payloads from pose/no-op/shape updates. The design goal is to generate once per track/archetype, update pose/material/ uncertainty every frame, and apply shape-parameter changes in place when the part topology is unchanged. When supplied by the SPPA observation bridge, the descriptor additionally preserves the observation identifier, covariance matrix, position/scale/yaw uncertainty estimates, and visual uncertainty policy. These fields are carried as auditable metadata; they do not make the visual uncertainty encoding a validated human-factors result. In the current generator, `descriptor_id` is a topology-cache identity: it excludes pose, timestamp, confidence, and observed color, which remain in `input_hash` or update fields. This separation is required for cheap per-frame updates; otherwise a pose update would look like a new mesh descriptor.

The third layer is an optional Unreal Engine 5.7 telemetry backend. The same obstacle payload can drive either curated asset actors or semantic proxy actors. The asset backend remains the default. The SPPA backend can ingest descriptor JSON, construct primitive components from the `parts` array, apply pose/no-op/shape update packets, and reject invalid JSON or mismatched descriptor identifiers without changing the current actor. This implements a runtime switch without forcing a migration away from curated assets.
The current resolver coverage regression checks 34 direct labels, including reviewed archetypes, conservative fallback contexts, and verifier-gated open-label recipes. A separate recipe-manifest contract checks 23 archetypes and 95 exact or keyword labels against the runtime generator. These checks verify resolver consistency and recipe roles; they are not evidence that the ontology is complete.

8 Preliminary Evaluation

All measurements in this section are preliminary operating-point tests. They are reported because they constrain the claim, not because they complete the validation of SPPA as a UAV/VR system.

8.1 Verifier-Gated Open-Label Probe

We evaluate the open-label route with 20 deliberately awkward labels that were not selected as the original paper demo classes. The set includes heavy equipment, vertical structures, containers, UAV-like objects, agricultural objects, and unsafe visual-artifact contexts. Table 2 shows the progression from the earlier contract-alignment probe to the current verifier-gated recipe implementation. The metric and tag-only conditions use the same labels; metric inputs provide declared dimensions, while tag-only inputs use template priors.

Table 2: Uncached open-label SPPA probe. Family/proxy means non-fallback runtime output. Verified recipes are accepted only after deterministic checks over required roles, fallback-role absence, primitive budget, and dimensions.

Condition	Fallback	Family/proxy	Verified recipes	Critical mismatches
Before alignment	12/20 raw, 13/20 norm.	8/20 raw, 7/20 norm.	0/20	7
After contract alignment	12/20	8/20	0/20	0
Verifier-gated recipes, metric	4/20	16/20	8/20	0
Verifier-gated recipes, tag-only	4/20	16/20	8/20	0

The remaining fallback cases in this probe are intentional: `dog statue`, `toy tractor`, `worker helmet`, and `cow silhouette` are visual-artifact or object-part contexts. This is a stronger claim than pure fallback but still narrower than arbitrary text-to-3D: SPPA accepts a generated proxy only when the recipe contract passes.

8.2 OBJ Construction Sanity Check

The original OBJ construction timing covered the initial seven-class prototype set and is retained only as a legacy construction/export sanity check. In that run, per-object export stayed below 0.1 s including Python startup and file export overhead, with meshes ranging from 488 vertices and 396 faces for `biker` to 856 vertices and 786 faces for `truck`. The current generator has since been extended to a broader finite-archetype resolver. These numbers are therefore implementation evidence for the OBJ construction path only, not current-ontology coverage, Unreal instancing timings, or end-to-end detection-to-render latency.

8.3 Benchmark Alignment and Generator Substitutability Stress Test

Table 3 is a substitutability stress comparison, not a SOTA ranking. A real SOTA ranking for image-to-3D would require the same detection crops or reference images, preferably with target meshes or structured human-preference scores, and should include contemporary high-fidelity systems such as TRELLIS.2, Pixal3D, Direct3D-S2, TripoSG/Tripo P1, Hunyuan3D 2.1, PartCrafter, and Rodin Gen-2.5 alongside fast feed-forward methods such as SF3D, SPAR3D, and TripoSR. SPPA would not be ranked as an image-to-3D asset generator in that protocol, because its intended input is a semantic detection state rather than a visual crop. The local stress test asks a narrower systems question: whether generators that were runnable in the local Windows workstation stack can replace a bounded runtime proxy under a portable-flight profile. Neural generators are expected to produce richer assets when they receive appropriate prompts or images; SPPA's question is persistence, updateability, and deterministic fallback from sparse telemetry.

All neural runs used the same RTX 5090 workstation with a 6 GB PyTorch CUDA allocator budget. This is a stress profile, not a physical GPU partition. It is motivated by 2026 laptop constraints: NVIDIA lists 8, 12, 16, and 24 GB memory configurations across RTX 50-series laptops [26], Epic recommends at least 8 GB graphics RAM for UE5 development [9], and the Steam May 2026 hardware survey reports large installed shares at 8, 12, and 16 GB [44]. The allocator limit follows PyTorch's documented memory-fraction mechanism [31].

Table 3: Preliminary July 2026 substitutability stress comparison over six object classes. Times are local wall-clock measurements under constrained input settings, not statistical or end-to-end UAV/VR validation.

Method	Input	Wall	Tris	VRAM
SPPA finite archetype	label/tag	0.0016 s med.	488–1,668	0 MB
Shap-E text $K = 16$	prompt/tag	2.0080 s	17,884–131,668	6,120 MB
Point-E text + SDF32	prompt/tag	6.4266 s	1,026–5,454	3,644 MB
TripoSR warm 128^3	RGBA crop	0.4604 s	19,284–31,012	2,028 MB
Hunyuan3D-2mini Turbo	RGBA crop	1.5042 s	332,020–1,698,032	6,092 MB

The defensible ranking from these data is therefore not a visual SOTA ranking. It is a task-fit ranking for the runtime contract that SPPA claims to satisfy: low-latency generation, bounded geometry, no neural GPU inference in the runtime loop, deterministic descriptors, and track-updateable actor semantics. Under that contract, Table 4 ranks the reproduced local methods by explicit pass/fail criteria. This ranking should be read together with Figure 2: neural generators can be visually richer when given suitable visual evidence, while SPPA is intentionally optimized for a different runtime representation problem.

Several contemporary image-to-3D candidates are intentionally excluded from the numeric table. Stable Fast 3D installed in a Python 3.10 environment but emitted no benchmark event before the 20 minute timeout in this pass. SPAR3D installation completed, but model loading

Table 4: SPPA runtime task-fit ranking derived from the local July 2026 generator stress runs. This ranking is for the UAV/VR semantic-proxy contract, not for image-to-3D visual SOTA. The six score criteria are: median generation below 33 ms, maximum mesh below 5k triangles, maximum torch reserved memory below 1 GB, no GPU inference, deterministic descriptor contract, and track-updateable actor semantics.

Method	Input	Score	Wall	Max tris	VRAM	Failed criteria
SPPA	normalized tag / gated observation	6/6	0.0031s	1,904	0MB	-
TripoSR warm	clean RGBA proxy crop	0/6	0.4604s	31,012	2028MB	lat., tris, mem, GPU, contract, update
Hunyuan3D-2mini Turbo	clean RGBA proxy crop	0/6	1.5042s	1,698,032	6092MB	lat., tris, mem, GPU, contract, update
Shap-E text K=16	prompt/tag	0/6	2.0080s	131,668	6120MB	lat., tris, mem, GPU, contract, update
Point-E text + SDF32	prompt/tag	0/6	6.4266s	5,454	3644MB	lat., tris, mem, GPU, contract, update

failed before inference because the required weights were gated. TRELLIS.2, Pixal3D, Direct3D-S2, TripoSG/Tripo P1, Hunyuan3D 2.1, PartCrafter, and Rodin Gen-2.5 were not executed under the frozen four-image local protocol. These are unresolved coverage gaps or reproducibility outcomes, not measured failures of output quality. Therefore the table is valid only as a bounded SPPA runtime substitutability stress test, not as a July 2026 visual image-to-3D SOTA ranking.

The fast-baseline boundary is narrower than the visual-SOTA boundary. For a fair runtime comparison against SPPA, a method must both be plausibly fast for the target operating point and have a successful local run under the same frozen input/resource protocol. Under that rule, TripoSR and Hunyuan3D-2mini Turbo are the reproduced image-conditioned rows in Table 3. Stable Fast 3D and SPAR3D remain the most important missing fast baselines: external sources report sub-second generation, but the local artifact contains only a timeout/no-event outcome for SF3D and a gated-weights outcome for SPAR3D. TRELLIS.2 reports fast low-resolution H100 timings, but it is not yet a reproduced portable-profile baseline here. The remaining contemporary methods are related visual-SOTA context, not fair fast-runtime rows without local timing provenance. The generated audit `benchmarks/results/sota_fast_method_eligibility.md` records this eligibility decision.

Figure 2 is a qualitative input-alignment audit across the six object cases. The first row shows the local crop/proxy image supplied to image-conditioned methods; it is not real detection ground truth. The final benchmark should replace this row with recorded detection crops, masks, bounding boxes, or a public benchmark reference set, and should report explicit quality metrics. The current image-conditioned methods were given clean proxy RGBA crops, while text-conditioned methods were run in speed-oriented settings. An input-provenance manifest and verifier mark all six current first-row images as synthetic proxy crops with zero ground-truth items. This prevents the figure from being re-labeled as a detector or ground-truth benchmark without new evidence. A separate synthetic YOLO reference manifest currently provides three 2D ground-truth bbox labels (`biker`, `cow`, and `tower`), but those legacy samples are bbox-only for this audit and do not provide readable crops for the six-class visual grid. The point of the figure is narrower: neural generators produce meshes with different fidelity/cost assumptions, while SPPA produces a bounded actor-like primitive proxy from the semantic label.

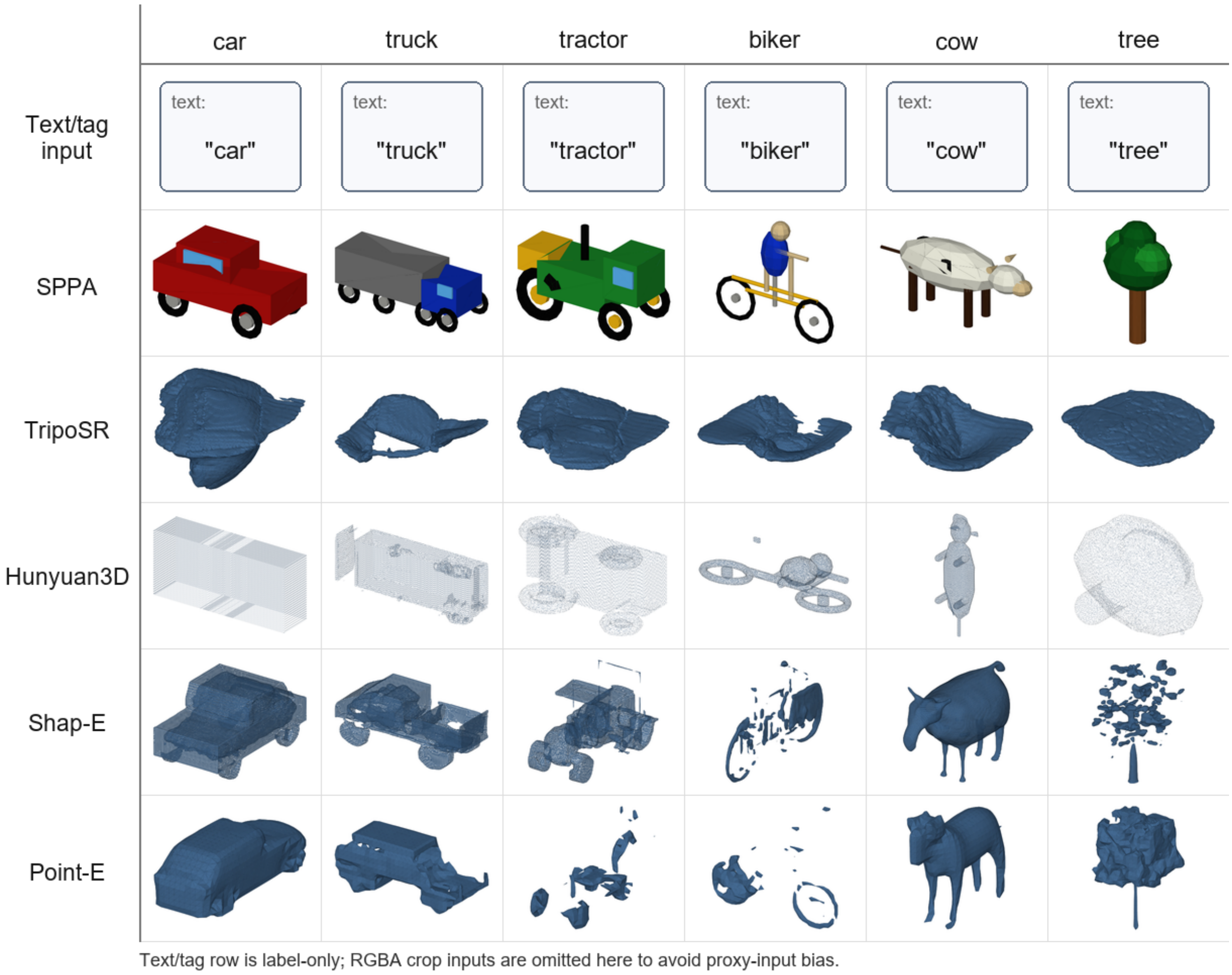

Figure 2: Input-aligned visual audit for the six local stress-test cases. The top row is deliberately text/tag only, not a proxy image and not ground truth. SPPA uses the current label-only finite-archetype prototype path; Shap-E and Point-E use text prompts/tags; TripoSR and Hunyuan3D use separate RGBA crop inputs that are omitted from this grid because proxy crops can visually resemble SPPA outputs. This is a qualitative audit artifact, not a SOTA ranking.

Figure 3 adds four real-image stress probes for the cyclist, electrical-tower, tractor, and tractor-with-trailer cases. These user-supplied crops are real input evidence, not 3D ground truth. The test is deliberately dual-input. In the image path, a YOLOE-26s open-vocabulary detector reads the real image and produces approximate semantic evidence. In the tag-only path, SPPA receives a semantic tag/text input without using the image crop. In both cases the semantic text is routed through a SPPA normalizer before proxy selection, so the paper does not depend on YOLOE producing an exact object name. The figure therefore shows a single SPPA output for each case rather than separate text-only and silhouette variants; the accompanying tables report how that proxy family is obtained from detector-only evidence, detector plus metric observation, and reviewed text. Detector masks, bboxes, and declared replay geometry are retained as evidence for audit, scale, pose, and uncertainty, but they are not shown as a separate competing SPPA method. For example, YOLOE only labels the tractor-with-trailer crop as `vehicle`. Detector-only normalization maps that weak label to a conservative heavy-vehicle proxy; when the same detector evidence is combined with its long metric footprint, SPPA refines it to `articulated_vehicle`. The reviewed text input `tractor_trailer` remains the more specific text-conditioned path and is reported separately. This is deliberately not a claim that the image path has identified the exact tractor or trailer type. Shap-E and Point-E receive tag/text prompts; TripoSR and Hunyuan3D receive the same low-quality image crops. The result should be read as an input-quality and semantic-normalization boundary check, not as a visual SOTA ranking.

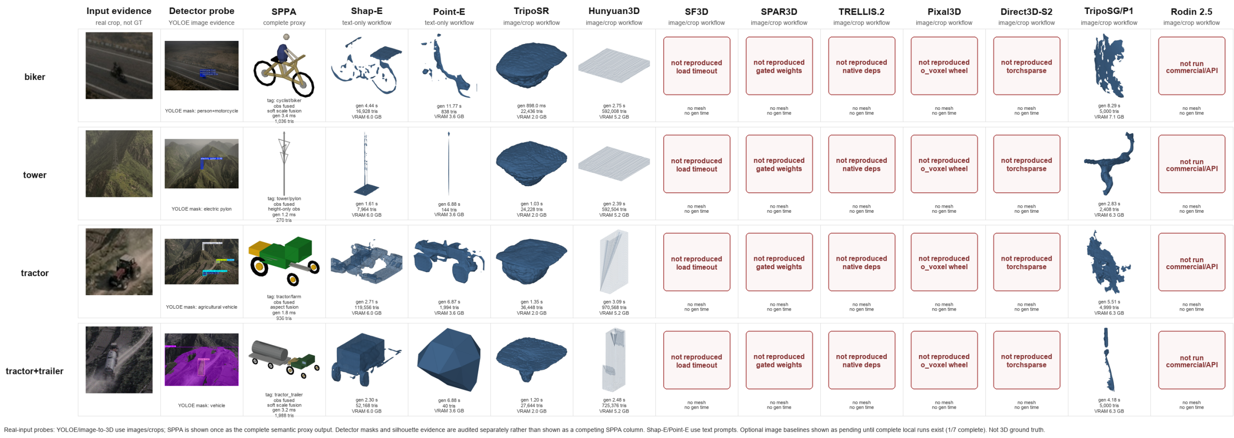

Figure 3: Real input probes for cyclist, electrical tower, tractor, and tractor-with-trailer cases. The first column is real input evidence, not ground truth. The detector path records YOLOE semantic evidence and reviewed 2D crop provenance. SPPA is shown once as the complete semantic proxy output, while YOLOE mask/bbox evidence and declared replay geometry are audited separately instead of being displayed as a second SPPA variant. Shap-E and Point-E use tag/text prompts; TripoSR and Hunyuan3D use image crops. This artifact demonstrates why the protocol must separate detector evidence, reviewed semantic text, crop annotations, and ground-truth 3D references.

Because the full comparison grid compresses the original frames, we add a zoomed detector-evidence audit in Figure 4. This makes the limits of the real-image probes visible: the cyclist, tower, and tractor-with-trailer cases are useful stress inputs, but their detector masks are moderate-confidence, small-object evidence rather than clean supervision. SPPA therefore records uncertainty and sends detector silhouette/bbox evidence through a class-aware constraint-fusion step before primitive generation; it does not treat the crop as a ground-truth mesh, mask, or metric measurement. In the real probes, this means vertical-structure height evidence may be kept with uncertainty even when the mask is weak, while noisy vehicle footprints can set a soft metric scale after their implausible length/width ratios are clamped by the semantic family prior. Unsafe image-derived yaw/pose is still withheld from the runtime descriptor. To keep the connector improvement aligned with the lightweight SPPA claim, the same four SPPA outputs are also checked by a runtime-budget regression: the current run totals 9.656 ms, with a worst single proxy of 3.383 ms, 1,988 triangles, 1,044 vertices, a 47.8 KB OBJ, and a 29.9 KB descriptor.

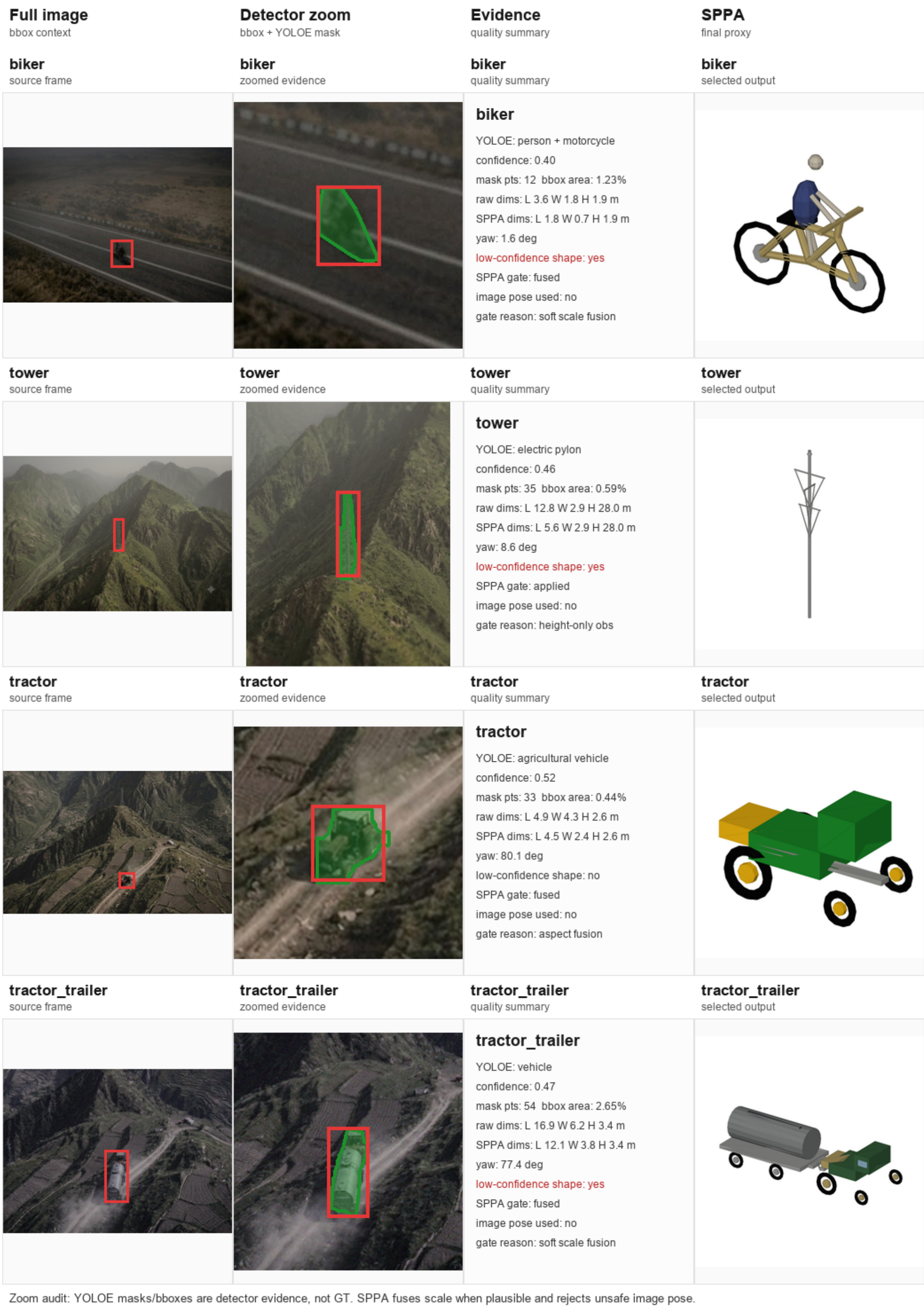

Figure 4: Zoomed real-input audit for the four user-supplied probes. Red boxes and green masks are YOLOE detector evidence. The evidence panel reports confidence, mask size, bbox area, inferred replay dimensions, yaw, and the SPPA low-confidence shape flag. The SPPA output is not split into separate text-only and silhouette variants; detector evidence is converted into one constraint-fused proxy. The evidence panel reports raw detector dimensions and the final SPPA dimensions, showing that the tractor's nearly square footprint is not copied but corrected through the vehicle aspect prior, while low-confidence image pose is not used. SPPA recipes include explicit connector primitives for semantically attached parts, such as bicycle frame tubes, tower braces, tractor axles, and tractor-trailer hitch/axle connectors.

Table 5: Real-input SPPA observation-fusion audit. Raw replay dimensions come from YOLOE mask/bbox projection under declared assumed-flight telemetry; SPPA dimensions are the selected constraint-fused proxy dimensions.

Case	Raw L/W/H (m)	SPPA L/W/H (m)	Gate	Pose	Tris	Boundary
biker	3.59 x 1.77 x 1.85	1.85 x 0.73 x 1.85	soft low-conf fusion	no	1036	scale only
tower	12.77 x 2.89 x 28.00	5.60 x 2.89 x 28.00	height-only fusion	no	270	scale only
tractor	4.95 x 4.32 x 2.60	4.52 x 2.38 x 2.60	soft aspect fusion	no	880	scale only
tractor_trailer	16.85 x 6.21 x 3.40	12.11 x 3.76 x 3.40	soft low-conf fusion	no	1904	scale only

Table 6: Real-probe semantic normalization. The image-side detector evidence is not treated as exact ground truth; SPPA maps it to a proxy family, optionally refining weak detector labels when metric observation supports a safer conservative shape.

Probe	YOLOE image evidence	Detector-only family	Detector+observation family	SPPA reviewed text
Cyclist road image	`person + motorcycle`	`two_wheeled_rider` → `biker`	`two_wheeled_rider` → `biker`	`biker`
Mountain tower image	`electric pylon`	`power_tower` → `vertical_structure`	`power_tower` → `vertical_structure`	`tower`
Mountain tractor image	`agricultural vehicle`	`farm_vehicle` → `farm_vehicle`	`farm_vehicle` → `farm_vehicle`	`tractor`
Tractor with trailer image	`vehicle`	`generic_vehicle` → `heavy_vehicle`	`articulated_vehicle` → `articulated_vehicle`	`tractor`+`trailer`

The dual-input benchmark is now frozen as an explicit artifact rather than a visual impression. The image track starts from YOLOE detector evidence on the real images and writes detector crops, confidence, and SPPA-normalized families. The tag/text track receives no image: SPPA, Shap-E, and Point-E use only the reviewed semantic phrase. Table 7 summarizes the bounded systems claim supported by this artifact. It remains separate from any full visual image-to-3D leaderboard because no 3D reference meshes or completed human-preference scores exist for these real images.

Table 7: Frozen real-image and tag/text dual-input benchmark for the bounded SPPA systems claim. The image track uses YOLOE detector evidence and crops for image-to-3D baselines; SPPA additionally reports an observation-refined detector family when geometry supports a safer proxy. The tag/text track uses the reviewed word or phrase for SPPA and text-conditioned baselines. This is not a full visual image-to-3D leaderboard.

Case	YOLOE image input	SPPA family path	Image-track methods	Tag/text methods
biker	`person + motorcycle` (0.40)	detector: `two_wheeled_rider` → `biker`; obs: `two_wheeled_rider` → `biker`; text: `biker` → `biker`	Hunyuan2-mini, TripoSR	Point-E, Shap-E, SPPA
tower	`electric pylon` (0.46)	detector: `power_tower` → `vertical_structure`; obs: `power_tower` → `vertical_structure`; text: `tower` → `tower`	Hunyuan2-mini, TripoSR	Point-E, Shap-E, SPPA
tractor	`agricultural vehicle` (0.52)	detector: `farm_vehicle` → `farm_vehicle`; obs: `farm_vehicle` → `farm_vehicle`; text: `tractor` → `tractor`	Hunyuan2-mini, TripoSR	Point-E, Shap-E, SPPA
tractor+trailer	`vehicle` (0.47)	detector: `generic_vehicle` → `heavy_vehicle`; obs: `articulated_vehicle` → `articulated_vehicle`; text: `tractor_trailer` → `tractor_trailer`	Hunyuan2-mini, TripoSR	Point-E, Shap-E, SPPA

Table 8 isolates the more ambitious image-side SPPA route. This route does not use the reviewed tag; it starts from YOLOE detector text and metric replay evidence. The important boundary is visible in the tractor-with-trailer case: the system refines `vehicle` to a conservative `articulated_vehicle`, not to the more specific reviewed `tractor_trailer` class.

Table 8: Detector-plus-observation SPPA route on frozen real probes. This route uses YOLOE semantic evidence plus replay metric dimensions, without the reviewed text tag, and remains bounded by the lightweight proxy budget.

Case	Detector+obs SPPA	Rule	ms	Tris	Boundary
biker	`biker` → `biker`	person+two-wheel	3.528	1036	conservative family
tower	`vertical_structure` → `vertical_structure`	power infrastructure	1.372	270	conservative family
tractor	`farm_vehicle` → `farm_vehicle`	farm vehicle	2.477	936	conservative family
tractor+trailer	`articulated_vehicle` → `articulated_vehicle`	long-footprint metric refinement	2.963	1848	conservative family

We also run a metric proxy replay on the same four real images (Table 11). This replay uses the real YOLOE detector outputs above, but injects declared UAV pose, AGL, camera mount, and family height priors as scenario parameters. It is therefore a real-image and real-detector test of the SPPA observation-to-descriptor path, not a measured flight-localization result. The replay now emits a `SPPA-OBS-0.1` identifier and descriptor uncertainty metadata for each case; those fields document the scenario-relative status rather than upgrading the replay to ground truth. The detector artifact now stores native YOLOE/Ultralytics mask polygons for the selected detections, so the replay can exercise the native detector-mask path rather than only a bbox or CV-derived silhouette proxy. These masks are still detector evidence, not ground-truth segmentations.

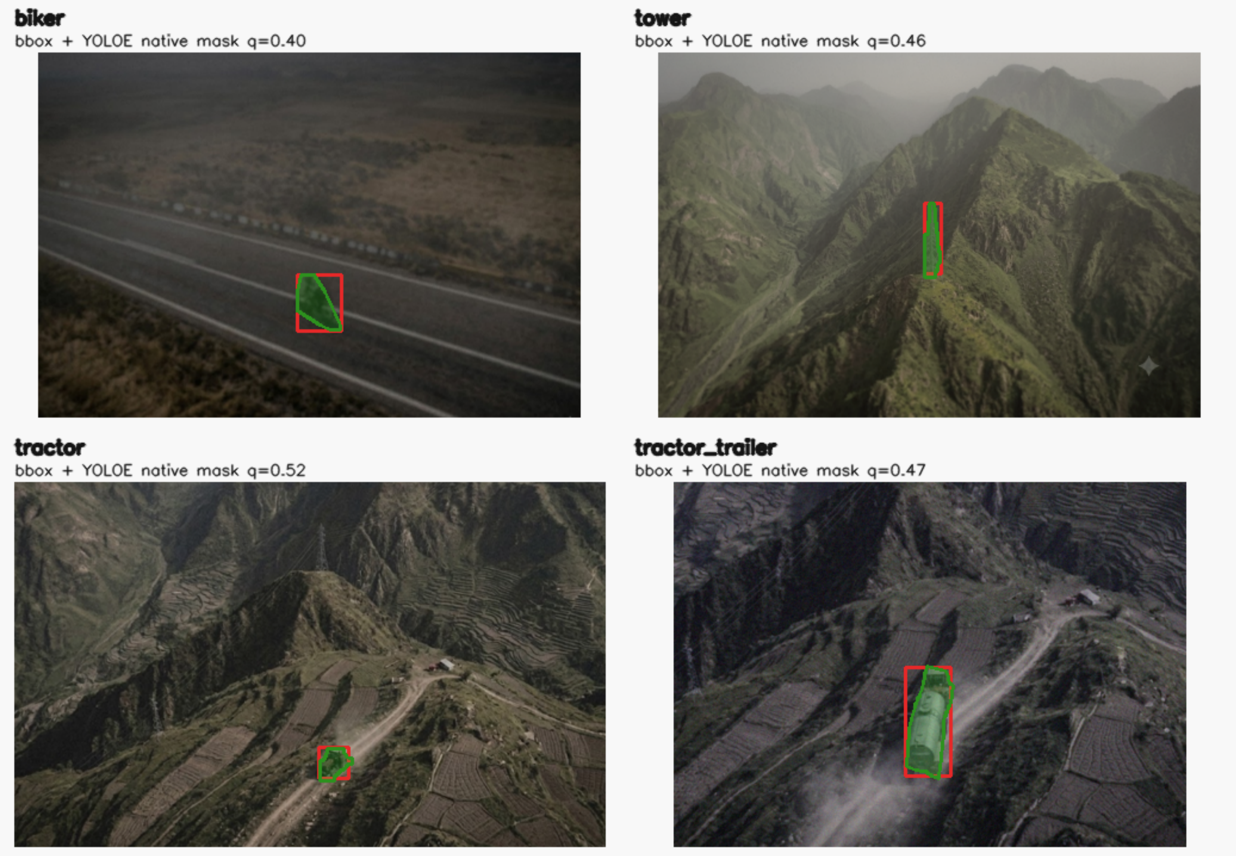

Figure 5: SPPA real-image replay evidence. Red boxes are the selected YOLOE or composite detector bboxes; green polygons are native YOLOE/Ultralytics mask contours used by the observation bridge. These polygons are detector evidence, not ground-truth masks.

To test the user's intended bridge from detector evidence to proxy parts without injecting class-specific part layouts, we added an experimental agnostic image-space primitive probe (Figure 7). The fitter receives only real crop pixels, the detector bbox, and unlabeled mask polygons; detector labels and reviewed SPPA tags are retained only for audit. It extracts generic cues such as round primitive pairs, coherent line structure, principal axes, and mask envelopes. The intended unit is therefore any detector crop, not a cyclist-specific or vehicle-specific rule: the same procedure is applied to all objects and can only report generic visual primitive evidence that is actually visible in the crop. On the four real images, the probe finds a strong round-pair cue for the cyclist and tractor, a coherent multi-line cue for the tower, and a multi-line cue for the tractor/trailer case while retaining raw circular primitives only as audit evidence. This supports a more ambitious image-to-primitive research path, but it is not yet a claim of universal part recovery or ground-truth 3D reconstruction. The current SPPA descriptor now consumes this evidence in a role-conditioned way (Table 9): generic round and line cues can annotate existing roles such as tires, frame elements, tower metal, attachments, or container details. The descriptor also stores a compact image-space geometry profile–for example round-pair axis/radius ratios or dominant line orientation–once per object, so the information is available for downstream pose/scale reasoning without duplicating it in every primitive. When no metric yaw, telemetry heading, track velocity, or mask-PCA yaw is available, SPPA also records an axial image-space orientation from the same profile. This orientation is explicitly marked as image-space and ambiguous; it is not a world yaw until a camera/ground projection module consumes it. These cues cannot create a new semantic class or add mesh topology. This is a deliberately bounded bridge from pixels to proxy structure.

Table 9: SPPA visual part evidence audit. Generic pixel cues from the detector crop annotate existing descriptor roles and store a compact image-space geometry profile; they do not change the semantic class or add mesh topology.

Case	Generic cue scope	Supported descriptor roles	ms	Tris	Boundary
biker	round-pair candidate	frame, hub, tire	3.383	1036	role support only
tower	multi-line candidate	tower metal	1.250	270	role support only
tractor	round-pair candidate	attachment, hub, tire	1.793	936	role support only
tractor+trailer	multi-line candidate	container detail, attachment	3.231	1988	role support only

The visual orientation field is also checked against the UAV metric projection branch (Table 10). This comparison is deliberately not used as a hidden correction step: raw pixel yaw and projected ground-footprint yaw are independent ambiguous axial cues with different error sources. The fair comparison therefore projects the visual axis endpoints (wheel-pair centers or the longest image-space line) through the same declared UAV camera model before comparing them with the footprint yaw. With this projection-aware gate the four real probes produce no strong divergence: two axes are aligned and two are weakly aligned under the declared replay geometry. SPPA still keeps the metric yaw tied to camera/flight projection and uses the projected visual axis only as support/gating evidence. When the gate is non-divergent, the descriptor selects the axial projected-footprint yaw in the declared replay frame; it does not treat the raw image-space axis as world yaw and does not claim ground truth. This makes the current system more conservative, but also more defensible: it shows how visible primitives can support pose reasoning only after the UAV projection model is in the loop. The descriptor schema and submission precheck enforce this as a conditional contract: whenever `visual_metric_yaw_consistency` is declared as evidence, the projected visual-axis gate, axial footprint-yaw selection, coordinate frame, ambiguity flag, and policy identifier must be present in the descriptor.

Table 10: SPPA visual–metric yaw gate on real image probes. Pixel-space visual axes are projected through the declared UAV replay camera model before comparison with the projected ground footprint. When the gate is non-divergent, SPPA selects the axial projected-footprint yaw; both projected quantities remain scenario-relative, not measured telemetry or ground truth.

Case	Visual axis	Footprint yaw	Δ	Agreement	Use in SPPA
biker	4.8°	1.6°	3.2°	aligned	selected axial footprint yaw
tower	35.8°	8.6°	27.2°	weakly aligned	selected axial footprint yaw
tractor	108.3°	80.1°	28.3°	weakly aligned	selected axial footprint yaw
tractor+trailer	101.9°	77.4°	24.5°	aligned	selected axial footprint yaw

Figure 6 shows the same bridge visually: the crop-level evidence is generic and unlabeled, the descriptor records only role support plus a compact image-space profile and, when available, a projection-gated axial footprint yaw; the proxy render remains the lightweight SPPA mesh. This is the current honest ambition boundary: SPPA uses visual evidence to condition parts and gate orientation fields that already belong to the normalized proxy, without claiming dense single-image reconstruction, detector-supervised part segmentation, or ground-truth world-frame yaw.

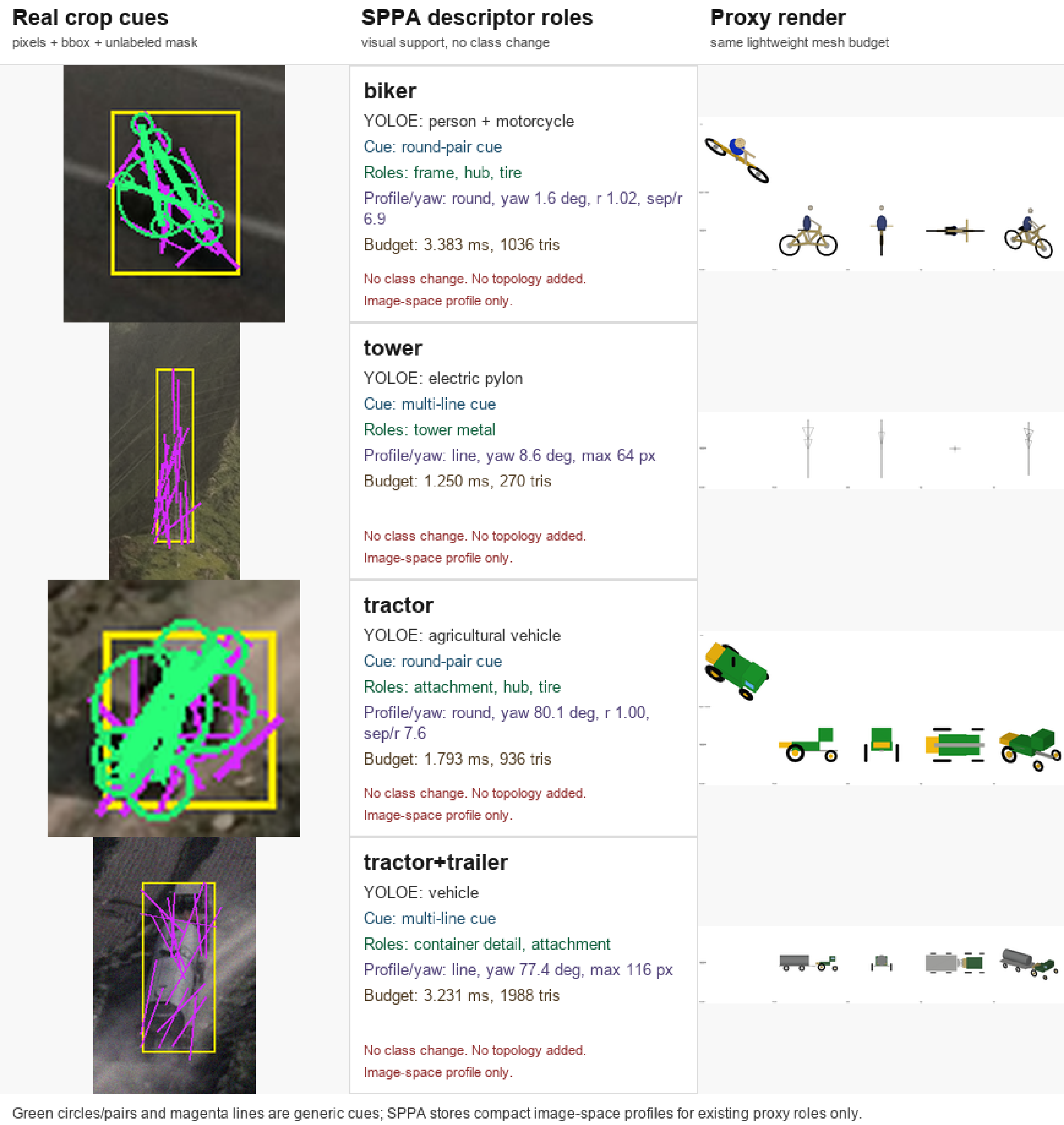

Figure 6: Visual part evidence bridge on the four real probes. Green circles/pairs and magenta lines are generic image-space cues from pixels, detector boxes, and unlabeled mask geometry. SPPA maps those cues only onto descriptor roles already present in the selected proxy and records a compact image-space geometry profile; when the visual–metric gate is non-divergent, the selected yaw is the declared-replay axial footprint yaw. The semantic class and mesh topology are unchanged.

A connector regression guards the main failure mode visible in naive primitive generators: disconnected pieces. In the four-probe run, the descriptor contains 15 explicit connectors for the cyclist proxy, six tower braces, two tractor axle connectors, and six tractor-trailer hitch/axle connectors; all connector endpoints are non-degenerate. This remains a lightweight semantic-connectivity claim, not a photorealistic reconstruction claim.

A companion label-invariance verifier mutates detector labels, reviewed SPPA tags, publication labels, detector model strings, and nested detection class IDs/names, then re-runs the fitter and hashes normalized geometry reports after removing audit-only label fields. All four reports remain unchanged, which supports the anti-label claim for this frozen replay without proving that the primitive cues themselves are correct. A second identity-invariance verifier renames the same rows to neutral unseen object IDs and again obtains unchanged geometry hashes after removing identity-only audit fields. This guards against case-name hardcoding. A third path-invariance verifier copies each real image to neutral filenames that contain no object words and obtains the same normalized geometry hashes. These tests leave primitive correctness to the visual evidence and controls, but they reduce the risk that the result is driven by hidden semantic or filename shortcuts. A fourth representation-invariance verifier reorders the detector masks, duplicates identical masks, and removes redundant native-mask payloads. The normalized geometry remains unchanged for all four real-image rows, after canonical polygon de-duplication and stable ordering by geometry. This last check is important for the intended "any detected object" setting: the bridge should react to visible geometry, not to incidental JSON ordering or repeated detector payloads. Finally, a combined side-channel verifier applies all non-geometric mutations at once: neutral case IDs, neutral image paths, adversarial labels and detector class fields, reversed and duplicated detector masks, and removal of redundant native masks. The normalized geometry again remains unchanged for all four real-image rows. This is a stronger anti-shortcut test than each mutation in isolation, although it still does not prove universal primitive correctness. We also added a horizontal mirror-equivariance verifier that mirrors each real image together with its bbox and unlabeled detector polygons. This initially exposed orientation sensitivity in the image-space primitive extractor, so the extractor now uses a symmetric circle/line candidate pass and a stricter line-structure gate. The primary geometry decisions remain stable for all four mirrored real-image rows; the report still records secondary audit warnings where Hough-line orientation statistics or weak round-pair evidence drift under mirroring. We also added a deterministic photometric-stability verifier that applies dark/low-contrast, bright/high-contrast, mild blur, and mild sensor-noise variants to the same four real images while keeping detector bboxes and unlabeled masks frozen. This stress test initially exposed a real weakness: small round cues could disappear under contrast/noise changes, and line-dominant structures could acquire false strong circular pairs. The current extractor therefore uses multiple photometric views for circular evidence (raw, CLAHE, normalized CLAHE, and smoothed normalized CLAHE) and contextually downgrades circular pairs inside clearly line-dominant structures unless round evidence is independently strong. After this change, all 16/16 photometric variants preserve the primary primitive family, with remaining audit warnings for confidence, line-count, and secondary pair-count drift. This is a regression guard against trivial illumination dependence, not a real-world illumination benchmark. We therefore added synthetic primitive controls: two round-pair positives, one line-structure positive, one blank-mask negative, and one texture negative. All five controls pass, including the requirement that the line-structure control has zero strong round-pair claims. A deterministic synthetic sweep further varies round-pair, elongated round-pair, line-structure, blank-mask, texture-negative, and single-circle-negative cases (72 cases total). The current fitter passes all 72 cases with synthetic primary-scope accuracy 1.000, strong round-pair precision/recall/F1 1.000/1.000/1.000, and line-structure precision/recall/F1 1.000/1.000/1.000. To reduce overfitting to a single deterministic grid, we also added a reproducible multi-seed synthetic fuzz test with four seeds and 240 randomized cases. This fuzz test initially exposed confusion between weak round-pair evidence and multi-line structures, motivating the context-aware strength gate described above. After this change the fuzz test passes all 240 cases with round-pair and line-structure precision/recall/F1 of 1.000/1.000/1.000. These are regression guards for primitive-cue logic, not evidence that real UAV imagery will achieve those rates.

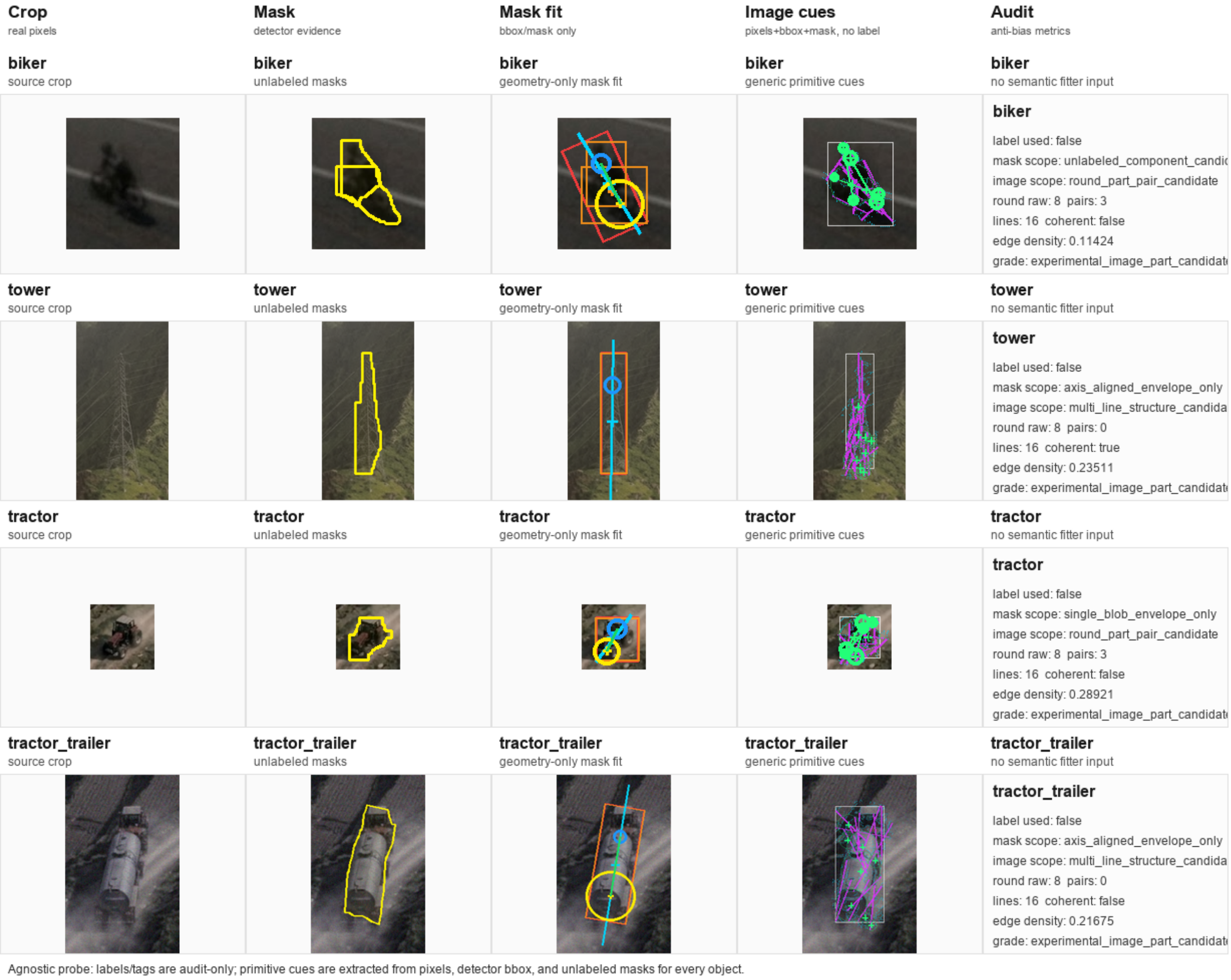

Figure 7: Experimental agnostic image-space primitive probe. The method uses real pixels, detector bboxes, and unlabeled masks for every object, while semantic labels are audit-only. Green circles indicate validated round-pair candidates when the geometry supports them; magenta lines show generic edge-line evidence. The right column records the anti-label audit and confidence grade for each case. This is an analysis probe for future image-conditioned part cues, not the runtime generator used for the SPPA outputs in Figure 3.

Table 11: Real-image SPPA metric replay using bbox plus native YOLOE mask polygon. Images and YOLOE detections are real local inputs; flight telemetry and height priors are declared replay parameters, so position and scale are scenario-relative rather than measured ground truth.

Case	YOLOE evidence	SPPA proxy	Scenario dims (m)	Local pose (m)
Cyclist	`person + motorcycle`	`biker`	$3.59 \times 1.77 \times 1.85$	$N = -3.67, E = -1.75$
Tower	`electric pylon`	`vertical_structure`	$12.77 \times 2.89 \times 28.00$	$N = -1.72, E = 0.56$
Tractor	`agricultural vehicle`	`farm_vehicle`	$4.95 \times 4.32 \times 2.60$	$N = -8.45, E = -10.96$
Tractor/trailer	`vehicle`	`heavy_vehicle`	$16.85 \times 6.21 \times 3.40$	$N = -2.38, E = -7.61$

A dedicated replay verifier checks this native-mask path. For the four real images, the selected YOLOE detections all include native mask polygons with 12, 35, 33, and 54 points respectively. The replay bridge projects those masks into candidate oriented footprints before the SPPA constraint-fusion gate decides which scale, footprint, and pose components are safe to carry into the selected runtime proxy. This verifies the detector-evidence data path and byte-budget handling; it still does not verify mask correctness against ground-truth segmentation, and the final descriptor may omit unsafe mask-derived pose.

The controlled SPPA contract benchmark isolates the difference between the old semantic-template prior and the metric-observation path. Synthetic masks are generated from declared object dimensions under a nadir pinhole camera, then compiled through the SPPA observation and descriptor contracts. This is a regression of contract behavior, not a real UAV segmentation benchmark. Under this controlled setting, the v1 semantic prior had summed dimension-error P50/P95 of 2.400/13.900 m; the v2 metric-observation path had 0.000/0.000 m. Mean improvement was 4.653 m, with descriptor build P50/P95 of 372.0/460.5 μs and a maximum of 1,692 triangles.

Table 12: SPPA controlled contract benchmark. v1 is the semantic template prior without metric observation; v2 uses synthetic metric mask evidence through `SPPA-OBS-0.1`. This supports scale-adaptation plumbing only, not real detector-mask quality or real flight calibration.

Case	Label	v1 err. (m)	v2 err. (m)	Improve (m)	v2 source
Cyclist long view	`cyclist`	0.470	0.000	0.470	mask-projected footprint
Tower tall proxy	`electric pylon`	13.900	0.000	13.900	mask-projected footprint
Tractor wide	`agricultural vehicle`	7.850	0.000	7.850	mask-projected footprint
Truck/trailer long	`truck`	2.400	0.000	2.400	mask-projected footprint
Cow close	`cow`	2.450	0.000	2.450	mask-projected footprint
Car small	`car`	0.850	0.000	0.850	mask-projected footprint

A second SPPA benchmark isolates the value of silhouette evidence relative to axis-aligned bboxes. We generate 162 synthetic rotated rectangles across three labels, three sizes, three aspect ratios, and six axial rotations under a known nadir camera, then run both paths through the same `SPPA-OBS-0.1` bridge. The bbox path over-estimates rotated footprints: summed dimensional error is 2.417/6.407 m at P50/P95 and axial-yaw error is 15.0/45.0°. The mask/silhouette path has 0.000/0.000 m error against the supplied synthetic polygon reference and preserves the polygon in 100% of generated descriptors. Non-axis-aligned median improvement is 2.951 m. This supports the contract claim that silhouette evidence prevents bbox inflation when such evidence exists; it is still not a claim that the real YOLOE replay above has ground-truth masks.

Table 13: SPPA bbox-vs-silhouette projection benchmark under known synthetic camera geometry. The mask path uses supplied synthetic polygons; real detector mask quality is outside this table.

Path	Cases	Dim. err. P50/P95 (m)	Yaw err. P50/P95	Descriptor evidence
Bbox-only projection	162	2.417/6.407	15.0/45.0°	axis-aligned bbox
Mask/silhouette projection	162	0.000/0.000	0.0/0.0°	polygon preserved in descriptor
Non-axis improvement	135	2.951/6.407	–	bbox inflation removed when polygon exists

These results narrow the argument. TripoSR and Hunyuan3D are better choices when the goal is high-fidelity asset generation from a usable image crop. Shap-E and Point-E demonstrate that text-conditioned meshes can be produced from the same tag path, but their outputs are not track-persistent proxy contracts. SPPA's measured advantage is different: no GPU inference, millisecond-scale local proxy creation, bounded primitive geometry, and per-track updateability. Whether that advantage matters to pilots is still unproven.

8.4 Role-Preservation Diagnostic

The main competitor set for SPPA is the set of operational representation alternatives in Table 1, especially curated assets and fast text/image-to-3D generators. Geometry-only displays are retained only as negative controls and lower-bound runtime references; construction timings for isolated boxes, capsules, or ellipsoids are therefore not reported as a result.

The implemented diagnostic asks whether SPPA preserves semantic part roles while adapting dimensions. For four synthetic vehicle variants, median dimension error was 0.3926 for fixed templates, 0.0000 for trivial global scaling, and 0.0223 for SPPA parametric vehicles. The global-scaling result is deliberately not a win condition: it fits the outer box by deforming all parts together, including cab and wheels. The SPPA diagnostic is therefore not a morphology benchmark. Its supported claim is narrower: role-assigned parts such as cargo, cab, windows, and wheels can follow different update rules instead of collapsing into a scaled copy of the same object. A dedicated short/long truck invariance check makes this concrete: increasing vehicle length from 5.2 m to 8.2 m kept cab scale delta at 0.0, tire scale delta at 0.0, increased cargo length by 2.925 m, and increased tire count from 6 to 8.

A calibrated-mask variant now exercises the same path from observed footprint evidence rather than manually supplied dimensions. Over 64 synthetic rectangle masks with explicit ground scale, the calibrated mask-to-shape benchmark derived metric length/width with 0.000/0.000 m P50/P95 summed dimensional error and kept the uncalibrated control path at `template_prior`. Local geometry build time was 442/480 μs at P50/P95, and descriptor build time was 297/321 μs at P50/P95. This verifies only the deterministic contract path *mask polygon + supplied ground scale* $\rightarrow$ *parametric parts*; it is not evidence of real UAV segmentation quality, camera calibration accuracy, or silhouette-to-part assignment under occlusion.

8.5 Descriptor and Unreal Runtime Tests

The descriptor/update benchmark generated descriptors for exact labels, keyword-resolved labels, and unsupported labels. Exact and keyword labels resolved to known finite archetypes; unsupported labels produced unknown fallback descriptors. In the replay benchmark, descriptor build with parts over 81,568 threshold-expanded rows had P50/P95 of 391.8/515.0 μs. The P99 was 671.2 μs, but the maximum descriptor-build observation was 360.050 ms. Scheduler decisions had P50/P95/P99 of 9.2/14.2/19.2 μs, and update-packet builds had P50/P95/P99 of 16.8/75.3/91.0 μs. Contract validation passed 453 descriptor and update-packet checks. These are Python contract measurements, not Unreal frame time; they support typical-case local contract cost, not a hard real-time guarantee.

A preliminary link-budget model over the same replay separates byte accounting from radio-link validation. If no-op packets are suppressed, measured SPPA JSON update traffic was 25.8–37.4 kB/s across scheduler thresholds; emitting every packet was 78.5–94.0 kB/s. A favorable modeled binary box stream was about 3.27 kB/s, while modeled compressed video at 2/5/10 Mbps corresponds to 250/625/1250 kB/s. This does not show that SPPA is the lowest-bandwidth semantic representation. It shows that SPPA is a richer actor representation with byte cost between sparse boxes and continuous video/mesh transfer under this replay model.

An additional adversarial resolver audit covered 12 compound or misleading labels. Visual-artifact contexts such as `toy truck`, `vehicle shadow`, `dog statue`, and `person poster` were forced to the unknown fallback; equipment-only labels such as `worker helmet` also fell back instead of spawning a person. Compound family labels such as `animal feed trailer`, `horse trailer`, `emergency vehicle`, `road sign`, and `field worker` resolved to reviewed family archetypes. This is a resolver-contract check, not a human-factors validation of fallback interpretation.

The offline recipe-review gate complements this resolver audit. It evaluates candidate language/ontology recipes against the finite primitive vocabulary, known material roles, runtime resolver result, triangle budget, and the rule that no language model may be called in runtime. In the current artifact, the gate accepted a `dump truck` candidate as a reviewed heavy-vehicle mapping, allowed an `unlisted obstacle` only as fallback, and rejected `person poster`, `delivery drone`, and `construction crane` candidates for false semantics, unreviewed topology/material roles, or runtime-LLM dependence. This supports the narrower claim that SPPA has a cache-admission guard for candidate recipes.

A mask-footprint benchmark then evaluated descriptor-level shape evidence over 162 synthetic rectangle masks: three lengths, three aspect ratios, six axial rotations, and three corner-jitter levels. Across clean non-axis-aligned masks, the AABB dimensional error was 50.568/109.808 px at P50/P95, while the PCA oriented footprint recovered 0.000/0.000 px dimensional error and 0.000/0.000 deg axial-yaw error. With jittered non-axis-aligned masks, AABB error was 51.957/114.235 px and oriented-footprint error was 6.615/12.389 px, with 0.158/0.666 deg yaw error. Axis-aligned masks can tie the AABB baseline, and jitter can make AABB slightly better for those aligned cases; the supported claim is therefore limited to extracting oriented image-space footprint evidence from supplied polygon masks, not real UAV silhouette recovery.

Unreal tests then exercised the native backend in three steps. First, an actor-level regression test loaded a real `SPPA-DESC-0.2` fixture with 12 parts, generated 12 static-mesh components, propagated material/evidence/ uncertainty tags, accepted valid pose updates, rejected mismatched descriptor updates, rejected invalid JSON without replacing the current proxy, and applied a matching-topology `shape_param_update` while preserving the same component names. Second, component replay used the same `obstacles` array payload shape as `/api/ui/data`. Third, a local-loopback HTTP replay exercised Unreal's HTTP polling branch with the same payload. A recorded-payload adapter then converted seven logged `obstacle_ingest` observations into common asset-compatible, SPPA emit-all, and SPPA changed-only payload streams. Unreal Editor-Cmd consumed those recorded streams through the same component entry point with asset, actor-proxy, and instanced-proxy backends. A later packaged executable run consumed the same streams with rendered frames. This is stronger than a payload-only adapter check, but it remains an internal short-stream replay; it is not live network telemetry, VR, phase-aligned GPU-profiler evidence, or operator validation.

The packaged evidence consists of synthetic density replay, HISM batching diagnostics, and a short recorded-payload replay. The baseline packaged replay, `20260702T122743Z`, used three repetitions, 30 warmup frames, and 120 measured frames per phase. At 100 synthetic obstacles, `semantic_proxy` used 432 components, about 208,864 estimated triangles, and 432 estimated draw calls; pose and shape frame P95 were 17.814 ms and 19.256 ms. This passed a 33.3 ms P95 desktop frame budget in that run but missed a 16.7 ms P95 budget, and phase-aligned GPU-profiler counters were not captured. The allowed claim is therefore limited to packaged desktop feasibility in specific measured cases, not VR readiness.

The recorded-payload packaged run, `20260703T024800Z`, replayed three externally generated JSONL streams through the same asset/proxy backend switch: common asset-compatible payloads, SPPA emit-all payloads, and SPPA changed-only payloads. All 27 applications were accepted. The HISM route had median active draw-group count 6 and frame P95 4.809 ms in this short replay, while the actor proxy had median component count 31 and frame P95 6.531 ms. This is useful integration evidence, but it is still a short internal replay without live network timing, phase-aligned GPU-profiler counters, headset execution, or operator evidence.

The HISM measurements are a separate opt-in diagnostic family, not a replacement for the baseline replay. In `20260702T215340Z`, a one-repetition run with shorter measurement windows, the HISM route reduced the 100-object component and estimated draw-call count from 432 to 11, and reduced `shape_param_update` action P95 from 20.736 ms to 13.270 ms. The 100-object shape-stream frame P95 improved from 42.903 ms to 33.626 ms, which was still above the 30 FPS P95 budget. The HISM update path was then changed to defer render-state dirty marking during batch instance updates and flush once per touched group. A longer HISM-only packaged run, `20260703T031246Z`, used three repetitions, 30 warmup frames, and 120 measured frames per phase at 100 objects. It reported 11 active draw groups, pose-stream frame P95 13.884 ms, shape-stream frame P95 29.715 ms, and shape-stream frame P99 35.021 ms, with 6 of 360 shape

Table 14: Selected Unreal evidence that constrains the SPPA claim. Values are preliminary and do not include VR headset, live-network, broad curated-asset validation, or phase-aligned GPU-profiler validation. Auxiliary regression checks are kept in the artifact package, not in this summary table.

Test	Result	Main boundary	Claim impact
Backend switch smoke	Same obstacle input selects assets, actor proxies, or HISM proxies; default remains assets	Editor smoke, not operator workflow	SPPA is optional, not a forced replacement
Editor-Cmd recorded replay	27/27 payload applications accepted across asset, actor-proxy, and HISM backends; SPPA emit-all P95 apply: 1.614/3.805/4.049 ms	component ingestion under Editor-Cmd/Null-RHI	recorded streams reach all backend routes
Packaged recorded replay `20260703T024800Z`	27/27 payload applications accepted; frame P95 assets/proxy/HISM: 16.551/6.531/4.809 ms; apply P95: 2.566/5.670/3.246 ms	one short replay, no live network, no phase-aligned GPU profiler, no headset	recorded streams run in packaged executable
Packaged synthetic 100-object replay	actor-proxy pose/shape frame P95: 17.814/19.256 ms	synthetic desktop obstacles; misses 60 FPS P95	desktop feasibility only, not VR readiness
Packaged HISM scaling `20260703T033655Z/033840Z`	11 active draw groups; 100-object no-CSV rerun shape frame P95: 37.126 ms; 250/500 shape P95: 94.018/186.113 ms	synthetic desktop obstacles; shape updates miss 30 FPS P95 at 100+ objects	batching helps draw pressure, but dense shape updates remain limiting
Packaged HISM partial updates `20260703T040602Z`	10% pose and 10% shape changed-track schedule; 500-object pose/shape frame P95: 16.528/29.787 ms with 11 draw groups	synthetic scheduler rate, not flight-derived; dense all-object updates still fail	partial-update contract supports changed-only scheduling, not dense all-track updates
Packaged cow/tower asset subset	project assets vs. SPPA shape frame P95: 6.789 vs. 22.445 ms	narrow two-label subset	curated assets remain better when available

frames above 33 ms. This was promising but not robust. A later dense sweep without Unreal CSV profiling, `20260703T033655Z`, tested 100, 250, and 500 objects with the same HISM backend. The 100-object shape-stream frame P95 was 38.657 ms, and an isolated 100-object no-CSV rerun, `20260703T033840Z`, reported 37.126 ms. At 250 and 500 objects, shape-stream P95 rose to 94.018 ms and 186.113 ms, respectively. The supported conclusion is therefore narrower: HISM batching keeps the proxy scene at 11 active draw groups across these synthetic densities and create-steady frames remain cheap, but stable 30 FPS P95 shape-update performance is not yet demonstrated even at 100 objects. This still does not support a 60 FPS, headset, live-network, or phase-aligned GPU-profiler claim.

The HISM path was then extended with an explicit `partial_obstacle_update` contract: unmentioned entities are retained, while mentioned entities replace their old instance keys before new parts are upserted. The backend verifier creates two entities, applies a partial shape-update to only one, and verifies 6/6 expected live/HISM instances after the untouched entity is preserved. A new packaged run, `20260703T040154Z`, confirmed that dense 100-object shape updates still fail (shape-stream P95 46.151 ms). With shape updates limited to 10% of objects but dense pose updates retained, shape-stream P95 stayed below 33.3 ms through 500 objects, but pose-stream P95 reached 52.474/104.502 ms at 250/500 objects. Under a changed-track schedule with both pose and shape updates limited to 10%, `20260703T040602Z` reported 500-object pose/shape frame P95 of 16.528/29.787 ms and no measured hitches above 33 ms. The allowed claim is therefore conditional: partial updates support a changed-only scheduler stress case, but the 10% rate is not yet derived from flight logs and does not prove dense all-object update performance.

Current HISM regressions expose `SEMANTIC_PROXY_INSTANCED` as a live opt-in backend, preserve local pose through the component route, reject mismatched descriptor/update packets, support group-level material and uncertainty tags, provide an opt-in collision policy, and support eight per-instance custom-data floats for resolved color, confidence, evidence source, and uncertainty. In an observed-color stress run at 100 objects, per-instance custom data reduced active HISM components/draw calls from 188 to 11 and reduced create/pose/shape action P95 from 27.792/9.669/18.303 ms to 18.608/7.050/15.351 ms, but did not improve the corresponding shape-stream frame P95 (37.725 vs. 38.136 ms). This supports HISM as an engineering direction for batching, not as completed dense-VR validation; phase-aligned GPU-profiler measurements and operational collision semantics remain open. The same backend verifier now includes a local georeferenced-pose regression: direct HISM payloads using `world_m` and `world_cm`, and component-routed payloads using both `SemanticProxy` and `SemanticProxyInstanced`, place a zero-centered descriptor at [1250,-400,120] cm with yaw 37 deg. This is only local-frame pose preservation, not evidence of live UAV georeferencing, camera calibration, Cesium accuracy, or VR runtime behavior.

A second packaged run adds a narrower real-asset baseline for the two labels with reviewed project static meshes in the repository, `cow` and `tower`. At 50 objects, `project_assets` used 50 static-mesh components and about 167,300 estimated triangles; `semantic_proxy` used 200 components and about 113,200 estimated triangles. The real-asset subset had lower P95 frame times for create, pose, and shape phases, with shape-stream P95 6.789 ms versus 22.445 ms for SPPA. This is not a failure of the SPPA claim: it shows that curated assets are preferable when an appropriate asset exists and loads cheaply. SPPA's intended gap is uncovered or mismatched semantic telemetry, not replacement of available high-quality assets.

9 Discussion

A strict interpretation is that SPPA currently has one defendable advantage and several unresolved weaknesses.

The advantage is runtime structure under sparse input. If the input is only a YOLO tag, track id, confidence, pose estimate, and rough dimensions, neural image-to-3D cannot use information that does not exist. A text-to-3D generator can hallucinate an asset, but it does not naturally produce a bounded, track-persistent, low-component Unreal actor with explicit yaw ambiguity, fallback reason, scale source, and update policy. SPPA does.

The weaknesses are also clear. Geometry is still based on finite archetypes. Open-label handling accepts any label but does not create correct geometry for any word. Mask/PCA footprint support reaches parametric geometry only when an explicit image-to-ground scale is supplied; otherwise it remains image-space descriptor evidence, and it has not been tested on real UAV imagery. Materials are procedural cues and semantic priors, with optional observed color applied only to predefined observable roles; they are not textures or evidence of visual truth. The artifact package includes a visual-material audit for the six paper SPPA renders, checking that OBJ materials are covered by manifests, rendered views are non-empty, and the renderer uses material manifests with depth-sorted face drawing. The actor-based Unreal implementation creates many primitive components, so draw calls and component counts can become limiting before triangles do. The new opt-in HISM route reduces this pressure substantially in the packaged benchmark and is now selectable as a live backend. Its basic `descriptor_id`/action contract and group-level material/color behavior have regression tests. Its query-collision policy is also regression-tested for confirmed versus tentative obstacles. A per-instance custom-data path now avoids extra HISM groups for heterogeneous color/confidence attributes, and a cooked material now reads the per-instance color/alpha channels in the opt-in path. The optimized HISM path now has packaged desktop evidence that draw-group pressure can stay low at 100–500 synthetic objects. Dense all-object updates remain too expensive: subsequent no-CSV-profiler reruns put the 100-object shape stream above the 30 FPS P95 target. The new partial-update contract improves the claim only under a changed-track scheduler: a synthetic 10% pose/shape schedule stayed below 33.3 ms P95 through 500 objects, while dense pose updates still failed at 250 and 500 objects. At 500 actors, the HTTP replay remains not real-time. These are not cosmetic limitations; they define the next research work.

SPPA also introduces a human-factors risk. A clean proxy can look more certain than the detector, georeferencing, yaw estimate, or scale estimate. SPPA therefore carries covariance-style uncertainty metadata from observation to descriptor, but the visual encoding must be tested with operators. Until then, an "uncertainty-marked generic fallback" means a specified display behavior, not a proven operational guarantee.

10 Threats to Validity

The main threats are methodological rather than cosmetic. First, mask and part-adaptation diagnostics use synthetic geometry, so they do not establish real UAV silhouette recovery. Second, the neural-generator comparison uses proxy RGBA crops or speed-oriented text prompts, so it evaluates substitutability under controlled inputs rather than general generator quality. Third, Unreal evidence comes from internal packaged and replay tests without a headset, phase-aligned GPU-profiler trace, dense mission scene, flight-derived changed-track rates, or live network loop. Fourth, the finite ontology can resolve labels into reviewed families, but it cannot produce correct geometry for arbitrary words. Fifth, SPPA reduces triangle counts in some cases but can increase component and draw-call pressure; the HISM diagnostic reduces that pressure in packaged replay and partial-update benchmarks, but does not yet establish dense VR readiness or live-mission scalability. Sixth, the YOLOE open-vocabulary probe is a preliminary detector source, not a proof of universal recognition. Its role in this paper is to supply imperfect compact semantic evidence that SPPA can normalize; target-drone latency, false-positive sweeps, and domain-specific fine-tuning remain required before flight use.

11 Claim Boundaries

The supported claim is narrow. Sparse semantic labels can be compiled into bounded proxy descriptors; those descriptors can be consumed by an optional Unreal backend that coexists with curated assets; and pose plus matching-topology shape updates can be applied on the tested paths without regenerating the actor topology. The current evidence also supports a semantic normalization layer that turns approximate detector labels into reviewed proxy families or explicit conservative fallback, plus internal role-preservation diagnostics for synthetic vehicle dimensions and synthetic rectangle-mask metadata.

The manuscript does not claim end-to-end UAV perception validation, universal object recognition, dense VR readiness, operator readability, network advantage, photorealistic asset generation, real-image silhouette recovery, or operational risk reduction. Those are future validation targets, not hidden conclusions.

12 Conclusion

SPPA is best framed as a deterministic semantic rendering contract for UAV digital twins, not as a general 3D generator and not as the flight system. Its current evidence supports a narrow claim: object-level telemetry can be compiled into low-polygon, part-based proxy descriptors; those descriptors can be ingested by an optional Unreal backend while preserving the existing asset backend; and pose plus matching-topology shape updates can be applied without regenerating components in the tested paths. The ambitious version of the system is therefore not "perfect 3D from anything"; it is YOLOE-style approximate detection, SPPA semantic normalization, and reviewed/fallback proxy families that stay lightweight, controllable, and explicit about uncertainty.

The paper is not yet evidence for a deployed UAV/VR system. The strongest next step is not more textual justification, but a live or recorded semantic-telemetry replay that feeds the same detections into both curated assets and SPPA, followed by packaged Unreal/VR profiling and a controlled operator study. Until those artifacts exist, SPPA should be judged as a specification plus preliminary runtime evidence for a semantic proxy representation layer.

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